NLP Labs & Dimensionality Reduction: PCA, t-SNE, UMAP

KAUST–Mawhiba IOAI Training

Week 3 · Day 14

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Table of Contents

1. Recap & Motivation
2. Why Dimensionality Reduction?
3. Principal Component Analysis (PCA)
4. t-SNE: Nonlinear Dimensionality Reduction
5. UMAP: Fast Nonlinear Reduction
6. Comparison & Summary

Recap: Days 12–13 in One Slide

- **Tokenization:** splitting text into units (words, sub-words via BPE)
- **Bag of Words & TF-IDF:** sparse document vectors
- **Word Embeddings:** dense vectors learned from data
  - Word2Vec, GloVe, fastText
  - Typically **100–300 dimensions** per word
- **Similarity:** cosine similarity measures semantic closeness
- **Word Arithmetic:** $\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}} \approx \vec{\text{queen}}$

Where We Left Off

We can now represent every word as a vector in $\mathbb{R}^{300}$:

	$\mathbf{d}_1$	$\mathbf{d}_2$	$\mathbf{d}_3$	$\mathbf{d}_4$	$\cdots$	$\mathbf{d}_{300}$
king	0.52	-0.31	0.18	0.94	$\cdots$	-0.07
queen	0.48	-0.27	0.22	0.89	$\cdots$	-0.05
cat	-0.61	0.73	-0.15	0.03	$\cdots$	0.44
dog	-0.55	0.68	-0.11	0.08	$\cdots$	0.39

Problem: We know "king" and "queen" are close, but...

How do you *visualize* 300-dimensional data?

Today's Roadmap

Morning: NLP Hands-On Labs

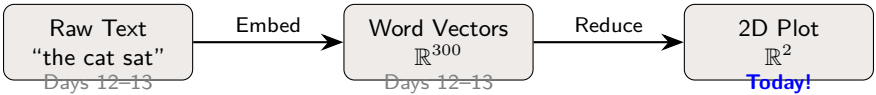
1. Tokenization lab — implement BPE, use HuggingFace tokenizers
2. Word embeddings lab — word analogies, cosine similarity, text search

Afternoon: Dimensionality Reduction

1. **PCA** — linear projection, find maximum variance directions
2. **t-SNE** — nonlinear, preserve neighborhoods
3. **UMAP** — fast nonlinear, preserve global structure

Goal: Go from 300D word vectors → a 2D scatter plot where similar words cluster together.

The Big Picture: From Words to Pictures



Why does this matter?

- **Visualization:** see clusters, outliers, relationships
- **Preprocessing:** reduce noise before classification
- **Speed:** fewer dimensions = faster training
- **Insight:** discover hidden structure in your data

Why **Dimensionality Reduction**?

What Is a "Dimension"?

Each **feature** in your data is one dimension:

Data Type	Features	Dimensions
Tabular (diabetes)	age, BMI, glucose, ...	$\sim$10
Image (28$\times$28 grayscale)	pixel intensities	784
Word embedding (GloVe)	learned dimensions	300
Document (BoW, 50K vocab)	word counts	50,000

High-dimensional data is everywhere in ML.

Humans can only see **2D** (or 3D) — we need to *compress* dimensions while keeping the important structure.

The Curse of Dimensionality

As dimensions increase, strange things happen:

- **Distances converge:** in high-D, the nearest and farthest neighbors are almost the same distance apart

$$\frac{d_{\max} - d_{\min}}{d_{\min}} \to 0 \quad \text{as } D \to \infty$$

- **Data becomes sparse:** 10 points in 1D $\to$ dense coverage; 10 points in 100D $\to$ nearly empty space

- **Overfitting risk:** more features than samples $\to$ model memorizes noise

Intuition

Imagine searching for your friend in a 1D hallway vs. a 100D hypercube. In high dimensions, *everywhere is far from everywhere else*.

Goals of Dimensionality Reduction

Reduce D features $\rightarrow$ d features ($d \ll D$), while:

1. **Preserving structure:**
   - Points that are close in high-D should stay close in low-D
   - Clusters should remain visible
2. **Removing noise:**
   - Many dimensions may carry redundant or noisy information
   - Keeping only the "important" directions cleans the signal
3. **Enabling visualization:**
   - Project to 2D or 3D for human inspection
   - Discover patterns, clusters, outliers

Two Families of Methods

Linear Methods

- New coordinates are **linear combinations** of original features
- Fast, deterministic, interpretable
- **Example:** PCA

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_D x_D$$

Nonlinear Methods

- Learn **curved** manifolds in the data
- Better at capturing complex structure
- **Examples:** t-SNE, UMAP

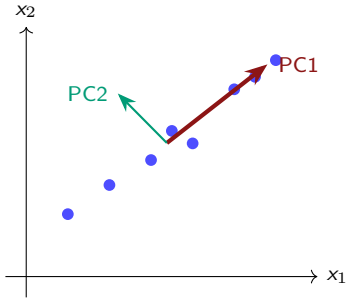
$$z = f(x_1, x_2, \ldots, x_D) \quad \text{(nonlinear } f\text{)}$$

Today we learn all three: PCA $\rightarrow$ t-SNE $\rightarrow$ UMAP

Principal Component Analysis: **PCA**

PCA: The Core Idea

Find the directions where data varies the most, then project onto those directions.



- **PC1:** direction of *maximum variance*
- **PC2:** perpendicular to PC1, captures remaining variance
- Projecting onto PC1 alone gives the best 1D summary

Analogy

PCA is like finding the "spine" of a data cloud — the line (or plane) along which the data is most spread out.

PCA: The Algorithm (Overview)

Given n data points in D dimensions, reduce to d dimensions ($d < D$):

1. **Center** the data: subtract the mean of each feature
2. **Compute** the covariance matrix $C \in \mathbb{R}^{D \times D}$
3. **Find** eigenvectors and eigenvalues of C
4. **Sort** eigenvectors by eigenvalue (largest first)
5. **Project** data onto the top d eigenvectors

Result: each data point is now a vector in $\mathbb{R}^d$.

Let's walk through each step with a concrete example!

Our Running Example: 5 Students, 2 Subjects

Dataset: 5 students, each with a Math and Science score.

Student	**Math** (x_1)	**Science** (x_2)
Alice	90	85
Bob	70	70
Carol	50	55
Dave	80	75
Eve	60	65

Question: Can we summarize each student with a *single number* instead of two? (2D $\rightarrow$ 1D)

We will do this step-by-step, then show Python does it in 3 lines.

Step 1: Center the Data

Subtract the mean of each feature so the data is centered at the origin.

Mean: $\bar{x}_1 = \frac{90+70+50+80+60}{5} = 70, \quad \bar{x}_2 = \frac{85+70+55+75+65}{5} = 70$

	Math	Science	Centered x_1	Centered x_2
Alice	90	85	$+20$	$+15$
Bob	70	70	0	0
Carol	50	55	-20	-15
Dave	80	75	$+10$	$+5$
Eve	60	65	-10	-5

Why center? PCA finds directions of variance *from the mean*. Without centering, the first component would just point toward the mean itself.

Step 2: Compute the Covariance Matrix

The **covariance matrix** measures how features vary *together*:

$$C = \frac{1}{n-1} X_c^T X_c \qquad \text{where } X_c \text{ is the centered data matrix}$$

$$X_c = \begin{pmatrix} 20 & 15 \\ 0 & 0 \\ -20 & -15 \\ 10 & 5 \\ -10 & -5 \end{pmatrix}$$

$$C = \tfrac{1}{4} \begin{pmatrix} 1000 & 700 \\ 700 & 500 \end{pmatrix} = \begin{pmatrix} 250 & 175 \\ 175 & 125 \end{pmatrix}$$

Reading the matrix:

- **Diagonal** $(250, 125)$: variance of Math, variance of Science
- **Off-diagonal** (175): covariance — Math and Science move together
- Positive covariance $\rightarrow$ high Math $\Leftrightarrow$ high Science

Quick Review: Variance and Covariance

- **Variance** of feature x:

$$\text{Var}(x) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

  Measures how spread out a single feature is.

- **Covariance** of features x and y:

$$\text{Cov}(x, y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

  Measures how two features change *together*.

- $\text{Cov} > 0$: both increase together $\text{Cov} < 0$: one increases as the other decreases

Our covariance $175 > 0$ confirms: students good at Math tend to be good at Science.

Step 3: Find Eigenvectors and Eigenvalues

We solve $C\mathbf{v} = \lambda\mathbf{v}$ to find:

- **Eigenvectors** $\mathbf{v}$: the *directions* of maximum variance
- **Eigenvalues** λ: the *amount* of variance in each direction

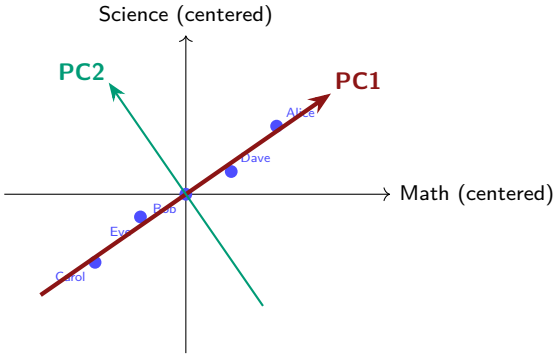
For our covariance matrix $C = \begin{pmatrix} 250 & 175 \\ 175 & 125 \end{pmatrix}$:

	Eigenvalue λ	**Eigenvector** $\mathbf{v}$
PC1	$\lambda_1 = 366.3$	$\mathbf{v}_1 = (0.82,\ 0.57)$
PC2	$\lambda_2 = 8.7$	$\mathbf{v}_2 = (-0.57,\ 0.82)$

Key observations:

- $\lambda_1 \gg \lambda_2$ — almost all variance is along PC1
- PC1 direction $(0.82, 0.57)$ is a mix of *both* subjects $\rightarrow$ "overall ability"

What Do Eigenvectors Look Like?



- **PC1** passes through the "spine" of the data cloud
- **PC2** is perpendicular — captures what PC1 misses
- Data is much more spread along PC1 than PC2

Step 4: Explained Variance Ratio

How much information does each component capture?

$$\text{Explained Variance Ratio of PC}_k = \frac{\lambda_k}{\sum_{j=1}^{D} \lambda_j}$$

	λ	**Variance %**	**Cumulative %**
PC1	366.3	$\frac{366.3}{375.0} = 97.7\%$	97.7%
PC2	8.7	$\frac{8.7}{375.0} = 2.3\%$	100%

PC1 captures 97.7% of all variance!

This means: we can compress 2 features into 1 and lose only 2.3% of the information.

In practice: keep enough PCs to capture $\geq 90\%$ of variance.

Multiply each centered data point by the PC1 eigenvector:

$$z_i = \mathbf{v}_1^T \cdot \mathbf{x}_{c,i} = 0.82 \cdot x_{c,1} + 0.57 \cdot x_{c,2}$$

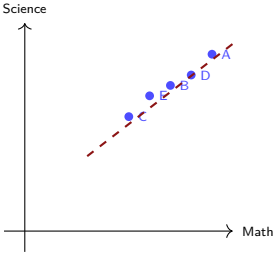
	Centered Math	Centered Sci	PC1 score z
Alice	$+20$	$+15$	$0.82(20) + 0.57(15) = \mathbf{+24.95}$
Bob	0	0	$0.82(0) + 0.57(0) = \mathbf{0.00}$
Carol	-20	-15	$0.82(-20) + 0.57(-15) = \mathbf{-24.95}$
Dave	$+10$	$+5$	$0.82(10) + 0.57(5) = \mathbf{+11.05}$
Eve	-10	-5	$0.82(-10) + 0.57(-5) = \mathbf{-11.05}$

The PC1 score is a single number that ranks students by "overall ability":

Carol (-25.0) Eve (-11.1) Bob (0) Dave $(+11.1)$ Alice $(+25.0)$

Visualizing the Projection

Original 2D data:



Projected to 1D (PC1):

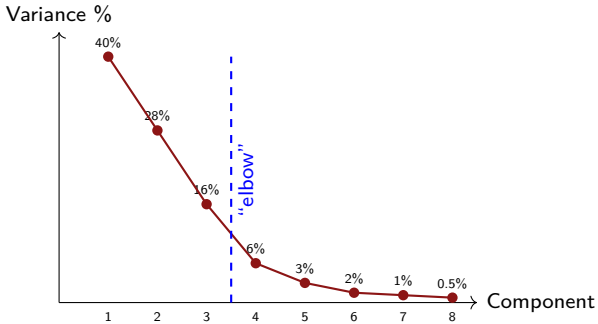
- Left: 2D scatter plot (2 numbers per student)
- Right: 1D projection (1 number per student)
- The **relative order and spacing** are preserved

This is exactly what PCA does with 300D word embeddings → 2D!

Choosing the Number of Components: The Scree Plot

In practice, you don't reduce from 2D to 1D — you reduce from **300D to 2D, 10D, 50D, …**

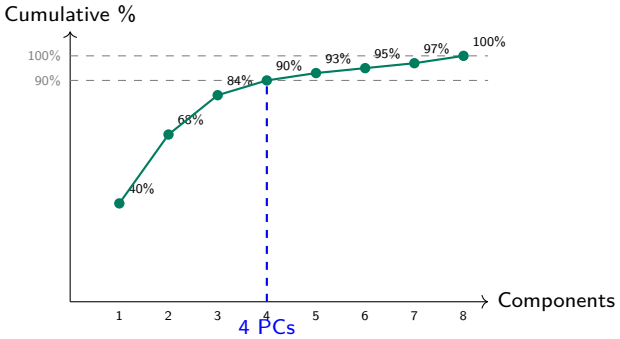
Scree plot: plot eigenvalues (or explained variance %) vs. component number.



Rule of thumb: keep components before the "elbow" where variance drops off

Cumulative Variance: The 90% Rule

An alternative view: plot **cumulative** explained variance.



Common choices:

- **Visualization:** $d = 2$ or $d = 3$ (always)
- **Preprocessing:** keep enough to reach 90–95% variance

PCA on Word Embeddings

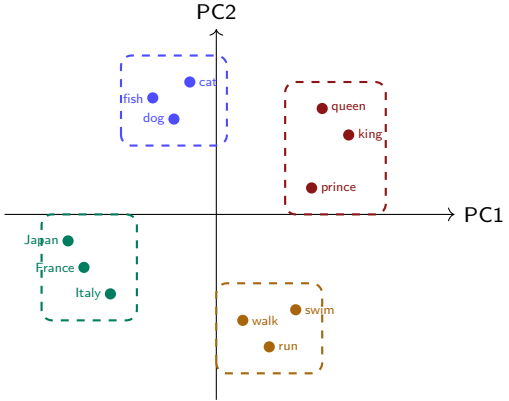
Real application: take 300D GloVe vectors $\rightarrow$ project to 2D for visualization.

1. Collect vectors for words of interest (e.g., animals, countries, verbs)
2. Center the data (subtract mean vector)
3. Compute the 300×300 covariance matrix
4. Find top 2 eigenvectors
5. Project each word vector to 2D
6. Plot and color by category!

What to expect

After PCA to 2D, you should see animals clustered together, countries

Example: PCA on 12 Word Vectors



PCA reveals semantic clusters! Royalty, Animals, Countries, Verbs group together.

PCA in Python: 3 Lines with Scikit-Learn

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
X_2d = pca.fit_transform(X_300d)

# X_300d: shape (n_words, 300)
# X_2d:   shape (n_words, 2)  -- ready to plot!
```

Useful attributes after fitting:

- `pca.explained_variance_ratio_` — how much variance each PC captures
- `pca.components_` — the eigenvectors (directions)
- `pca.singular_values_` — related to eigenvalues

PCA in Python: Full Example

```python
import numpy as np
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Example: 5 students, 2 subjects
X = np.array([[90,85],[70,70],[50,55],[80,75],[60,65]])
names = ['Alice','Bob','Carol','Dave','Eve']

# Fit PCA (2D -> 1D)
pca = PCA(n_components=1)
X_1d = pca.fit_transform(X)

print(f"Explained variance: "
      f"{pca.explained_variance_ratio_[0]:.1%}")
# Output: Explained variance: 97.7%

# Scree plot for higher-D data:
pca_full = PCA().fit(X_300d)
plt.plot(np.cumsum(pca_full.explained_variance_ratio_))
plt.xlabel('Number of Components')
plt.ylabel('Cumulative Variance Explained')
plt.axhline(y=0.9, color='r', linestyle='--')
plt.show()
```

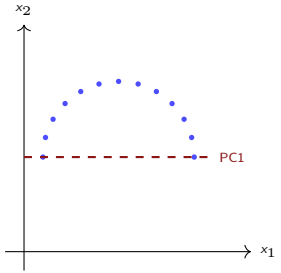
PCA: Strengths

- **Fast:** closed-form solution (eigen-decomposition), scales to large D
- **Deterministic:** same input $\rightarrow$ same output (no randomness)
- **Interpretable:** each PC is a weighted sum of original features
- **Preserves global structure:** distances and variances are maintained
- **Great for preprocessing:** reduce features before training a classifier
- **Explained variance:** tells you exactly how much information you keep

PCA: Limitations

- **Linear only:** can't capture curved or twisted structures
- **Assumes variance = importance:** high variance direction might be noise
- **Visualization limited:** 2 PCs may capture too little variance for complex data

When PCA fails:
If data lies on a curve (manifold), PCA projects onto a straight line and merges clusters.



Data on a curve → PCA projection loses the structure.

This is exactly why we need **nonlinear methods** like t-SNE and UMAP!

PCA: Key Takeaways

1. **PCA finds orthogonal directions of maximum variance**
2. The algorithm: center $\rightarrow$ covariance $\rightarrow$ eigenvectors $\rightarrow$ project
3. Use the **scree plot** or **90% rule** to choose how many components
4. **Strengths:** fast, deterministic, interpretable, good for preprocessing
5. **Weakness:** linear — misses curved/nonlinear structure
6. In Python: `PCA(n_components=k).fit_transform(X)`

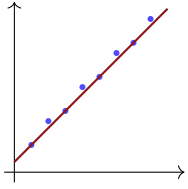
Up next: t-SNE — a nonlinear method that preserves *local neighborhoods*.

t-SNE: **Nonlinear Dimensionality Reduction**

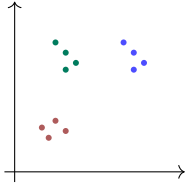
Why Not Just Use PCA?

PCA is great when structure is **linear**, but real data often isn't.

PCA works well:



PCA fails:



With clusters in high-D:

- PCA might project two separate clusters onto the *same* region
- We need a method that preserves **which points are neighbors**

t-SNE: The Core Idea

t-SNE = **t**-distributed **S**tochastic **N**eighbor **E**mbedding

- **Goal:** if two points are close in high-D, keep them close in low-D
- **Approach:**
  1. Measure how "similar" every pair of points is in high-D
  2. Place points randomly in 2D
  3. Iteratively move points in 2D until neighbors match the high-D layout
- **Result:** a 2D map where *neighborhoods are preserved*

Analogy

Imagine you have a social network with 1000 people. You want to place them on a map so that friends are close together. That's what t-SNE does with data points.

How t-SNE Works (Intuition)

Step 1: Measure similarities in high-D

For every pair of points, compute a **similarity score**:

- Close points → **high** similarity
- Far points → **low** similarity (near zero)

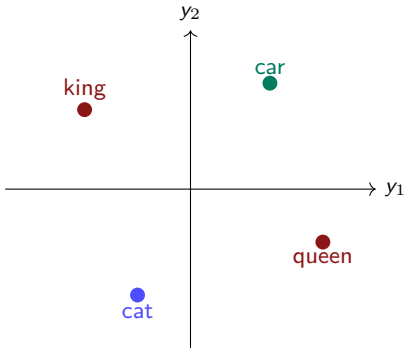
Example with 4 word vectors:

Similarity	king	queen	cat	car
king	—	**high**	low	low
queen	**high**	—	low	low
cat	low	low	—	low
car	low	low	low	—

King and queen are close in 300D → high similarity score.

Cat and car are far from everything → low similarity scores.

Step 2: Start with random positions in 2D



Random positions — king and queen are far apart!

The initial placement is random and wrong.

t-SNE will **iteratively rearrange** points to fix this.

How t-SNE Works (Intuition)

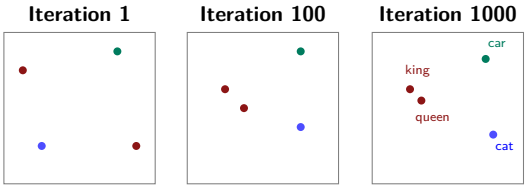
Step 3: Iteratively move points to match the similarities

- Points that should be close (high similarity) are **attracted**
- Points that should be far (low similarity) are **repelled**
- Repeat for hundreds of iterations until the layout stabilizes

Think of it like a **physics simulation**:

- Similar points have *springs* pulling them together
- Dissimilar points have *repulsive forces* pushing them apart
- The system settles into an equilibrium

Watching t-SNE Converge



- King and queen are pulled **together** (high similarity in 300D)
- Cat and car stay **apart** (low similarity in 300D)
- The exact positions don't matter — only **relative closeness**

The Perplexity Parameter

Perplexity controls how many neighbors each point pays attention to.

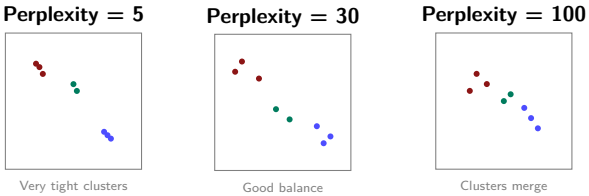
Think of it as: "How many close friends does each point have?"

- **Low perplexity** (5–10): each point only considers its *nearest few* neighbors $\rightarrow$ very tight, small clusters
- **Medium perplexity** (30–50): a good balance between local detail and overall shape
- **High perplexity** (100+): each point considers *many* neighbors $\rightarrow$ clusters may merge

Rule of Thumb

Typical range: **5–50**. Default in scikit-learn: **30**. Try a few values and compare the plots.

Effect of Perplexity: Same Data, Different Views



- Too low: fragmented, spurious sub-clusters
- Too high: clusters blend together, lose local detail
- **Always try multiple values** and look at the results!

t-SNE in Python

```
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

tsne = TSNE(n_components=2, perplexity=30,
            random_state=42)
X_2d = tsne.fit_transform(X_300d)

plt.scatter(X_2d[:, 0], X_2d[:, 1],
            c=categories, cmap='tab10')
plt.title('t-SNE of Word Embeddings')
plt.show()
```

Key parameters:

- `perplexity` = 30 (default, try 5–50)
- `n_iter` = 1000 (default, increase if not converged)
- `random_state` = 42 (for reproducibility)

t-SNE: Important Caveats

1. **Non-deterministic:** different random seeds → different plots. Always set `random_state`!
2. **Distances between clusters are meaningless:** only the points *inside* each cluster are reliable
3. **Axes are meaningless:** no "PC1 explains 40% variance"
4. **Cannot add new points:** to embed a new point, you must rerun on the entire dataset
5. **Cluster sizes can be misleading:** t-SNE may inflate small clusters and compress large ones
6. **Slow for large datasets:** works well up to ∼50K points

t-SNE is for **visualization only**, not as input to another model.

What t-SNE Preserves (and Doesn't)

Property	Preserved?	Reliable?
Points in the same cluster	✓	Yes
Local neighborhood (nearest neighbors)	✓	Yes
Distance between clusters	✗	No
Cluster sizes / shapes	✗	No
Global layout / orientation	✗	No
Explained variance	✗	N/A

Bottom line: if two points are close in a t-SNE plot, they are likely similar. If two *clusters* are far apart, that may or may not be meaningful.

t-SNE: Key Takeaways

1. **t-SNE preserves local neighborhoods** — similar points stay close
2. Works by iteratively moving 2D points until they match high-D similarities
3. **Perplexity** controls how many neighbors to consider (try 5–50)
4. **Great for:** visualizing clusters in word embeddings, images, etc.
5. **Limitations:** slow, non-deterministic, distances between clusters are meaningless
6. In Python: `TSNE(perplexity=30).fit_transform(X)`

Up next: UMAP — faster, preserves more global structure.

UMAP: **Fast Nonlinear Reduction**

t-SNE Is Great, But...

- **Slow:** $O(n^2)$ — minutes for 10K points, hours for 100K+
- **No global structure:** distances between clusters are meaningless
- **Can't add new points:** must rerun on entire dataset
- **Non-deterministic:** hard to reproduce results

UMAP (Uniform Manifold Approximation and Projection) addresses all of these.

McInnes, Healy & Melville (2018) — now the most popular visualization method.

UMAP: The Core Idea

Like t-SNE, UMAP preserves neighborhoods. But the math is different:

1. **Build a graph** in high-D:
   - Connect each point to its k nearest neighbors
   - Edge weights = distance-based similarity
2. **Optimize a layout** in 2D:
   - Find a 2D arrangement where the graph structure is preserved
   - Uses cross-entropy instead of KL divergence

Key Difference from t-SNE

t-SNE considers *all* pairwise distances ($O(n^2)$). UMAP only considers *nearest neighbors* ($O(n \log n)$) — much faster!

UMAP's Two Key Parameters

`n_neighbors` (default: 15)

How many neighbors to consider for each point.

- **Small** (5): very local, tight clusters
- **Medium** (15): balanced view
- **Large** (50+): broader, more global structure

Similar to t-SNE's perplexity.

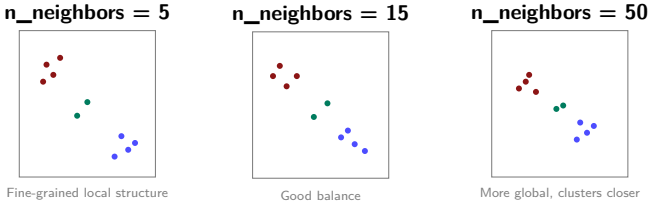
`min_dist` (default: 0.1)

Minimum distance between points in 2D.

- **Small** (0.0): points can overlap, very tight clusters
- **Medium** (0.1): slight spacing
- **Large** (0.5+): spread out, more uniform

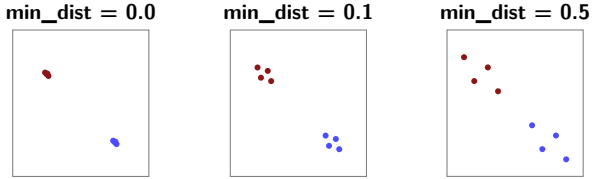
No equivalent in t-SNE.

Effect of n_neighbors



- Small `n_neighbors`: emphasizes local clusters (may fragment)
- Large `n_neighbors`: preserves global relationships (may merge)

Effect of `min_dist`



- `min_dist = 0.0`: tight clusters, good for cluster identification
- `min_dist = 0.5`: spread out, better for seeing internal structure

UMAP vs. t-SNE: Speed Comparison

Dataset Size	t-SNE	UMAP
1,000 points	$\sim$5 sec	$\sim$2 sec
10,000 points	$\sim$2 min	$\sim$15 sec
100,000 points	$\sim$30 min	$\sim$2 min
1,000,000 points	hours / infeasible	$\sim$15 min

Why is UMAP faster?

- t-SNE: computes *all* n^2 pairwise distances
- UMAP: only computes *k-nearest neighbor* distances ($n \log n$)
- UMAP uses stochastic gradient descent with negative sampling

UMAP Preserves More Global Structure

t-SNE:

- Local neighborhoods: ✓
- Cluster separations: ✗
- Relative cluster positions: ✗

Clusters may appear in arbitrary positions. The distance *between* clusters is meaningless.

UMAP:

- Local neighborhoods: ✓
- Cluster separations: ✓ (partially)
- Relative cluster positions: ✓ (partially)

Clusters that are related in high-D tend to be *closer* in the UMAP plot.

Example

In a UMAP of word embeddings: "king" and "queen" clusters would be closer to each other than to "cat" and "dog" clusters — t-SNE doesn't guarantee this.

UMAP in Python

```
# pip install umap-learn
import umap
import matplotlib.pyplot as plt

reducer = umap.UMAP(n_neighbors=15,
                    min_dist=0.1,
                    random_state=42)
X_2d = reducer.fit_transform(X_300d)

plt.scatter(X_2d[:, 0], X_2d[:, 1],
            c=categories, cmap='tab10', s=10)
plt.title('UMAP of Word Embeddings')
plt.show()
```

Key parameters:

- `n_neighbors` = 15 (default, try 5–50)
- `min_dist` = 0.1 (default, try 0.0–0.5)

UMAP: Bonus Features

1. **Transform new points:**
   - After fitting, call `reducer.transform(X_new)`
   - t-SNE cannot do this — must rerun on everything
2. **Supervised UMAP:**
   - Pass labels: `reducer.fit(X, y=labels)`
   - Uses label information to improve cluster separation
3. **Higher-dimensional output:**
   - Not just 2D — can project to 10D, 50D for downstream ML
   - Use as a preprocessing step (like PCA, but nonlinear)

UMAP is both a **visualization tool** and a **feature extraction method**.

UMAP: Key Takeaways

1. **UMAP builds a nearest-neighbor graph** and optimizes a 2D layout
2. Two main parameters: `n_neighbors` (locality) and `min_dist` (density)
3. **Much faster** than t-SNE ($O(n \log n)$ vs. $O(n^2)$)
4. **Preserves more global structure** than t-SNE
5. **Can transform new data** without refitting
6. Install via `pip install umap-learn`

PCA vs. t-SNE vs. UMAP

Head-to-Head Comparison

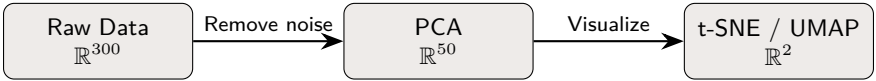
	PCA	t-SNE	UMAP
Type	Linear	Nonlinear	Nonlinear
Speed	Very fast	Slow ($O(n^2)$)	Fast ($O(n \log n)$)
Deterministic?	✓ Yes	✗ No	✗ No
Local structure	Partial	✓ Excellent	✓ Excellent
Global structure	✓ Yes	✗ No	Partial
Interpretable axes	✓ Yes	✗ No	✗ No
Explained variance	✓ Yes	✗ No	✗ No
Add new points	✓ Yes	✗ No	✓ Yes
Best for	Preprocessing	Visualization	Both

When to Use Which Method

- **Use PCA when:**
  - You need *interpretable* components ("what drives the variation?")
  - You want a *preprocessing* step before a classifier
  - Speed matters or dataset is very large
  - You need to explain how much information is retained
- **Use t-SNE when:**
  - You want to *visualize clusters* in 2D
  - Dataset is small–medium ($<$ 50K points)
  - You want the most detailed local structure
- **Use UMAP when:**
  - You want fast visualization for *large* datasets
  - Global relationships between clusters matter

The Practical Pipeline: PCA + t-SNE/UMAP

The standard workflow combines methods:



Why use PCA first?

1. **Denoise:** remove dimensions that are mostly noise
2. **Speed:** t-SNE on 50 features is much faster than on 300
3. **Better results:** t-SNE/UMAP work better on cleaner data

Best practice

`PCA(n_components=50)` → then `TSNE()` or `UMAP()`. This is what most practitioners do!

Complete Pipeline in Python


```python
import numpy as np
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import umap
import matplotlib.pyplot as plt

# Load word embeddings: shape (n_words, 300)
X = load_word_vectors(words)

# Step 1: PCA to 50D (denoise + speed up)
pca = PCA(n_components=50)
X_50 = pca.fit_transform(X)
print(f"Variance kept: "
      f"{sum(pca.explained_variance_ratio_):.1%}")

# Step 2a: t-SNE to 2D
tsne = TSNE(perplexity=30, random_state=42)
X_tsne = tsne.fit_transform(X_50)

# Step 2b: UMAP to 2D
reducer = umap.UMAP(n_neighbors=15, min_dist=0.1)
X_umap = reducer.fit_transform(X_50)
```

Plotting Side by Side

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# PCA plot
axes[0].scatter(X_50[:, 0], X_50[:, 1],
                c=colors, s=20)
axes[0].set_title('PCA (first 2 PCs)')

# t-SNE plot
axes[1].scatter(X_tsne[:, 0], X_tsne[:, 1],
                c=colors, s=20)
axes[1].set_title('t-SNE (perplexity=30)')

# UMAP plot
axes[2].scatter(X_umap[:, 0], X_umap[:, 1],
                c=colors, s=20)
axes[2].set_title('UMAP (n_neighbors=15)')

# Annotate words
for i, word in enumerate(words):
    for ax in axes:
        ax.annotate(word, (X_method[i,0], X_method[i,1]),
                    fontsize=6)
plt.tight_layout()
plt.show()
```

Day 14: Key Takeaways

1. **Dimensionality reduction** compresses high-D data to low-D while keeping structure
2. **PCA** is linear: finds max-variance directions, fast, interpretable, good for preprocessing
3. **t-SNE** is nonlinear: preserves neighborhoods, great for visualization, slow
4. **UMAP** is nonlinear: fast like PCA, visual quality like t-SNE, preserves global structure
5. **Best practice:** PCA (denoise) $\rightarrow$ t-SNE/UMAP (visualize)

In your lab: apply all three methods to word embeddings and see animals, countries, and verbs form distinct clusters!

Next: Day 15 — CV Competition

Day 16: Clustering — "You saw the clusters. Now find them algorithmically."

Quick Reference Card

Method	Python Code	Key Parameter
PCA	`PCA(n_components=k).fit_transform(X)`	`n_components`
t-SNE	`TSNE(perplexity=30).fit_transform(X)`	`perplexity` (5–50)
UMAP	`UMAP(n_neighbors=15).fit_transform(X)`	`n_neighbors`, `min_dist`

Imports:

```
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import umap   # pip install umap-learn
```

Typical pipeline:

```
X_50 = PCA(50).fit_transform(X_300)
X_2d = UMAP().fit_transform(X_50)
```

Credits

KAUST Academy

King Abdullah University of Science and Technology

Dimensionality reduction content partially adapted from

KAUST Academy Unsupervised Learning course materials

KAUST–Mawhiba IOAI 2026 Training Program